

Analysis of the ChickWeight dataset

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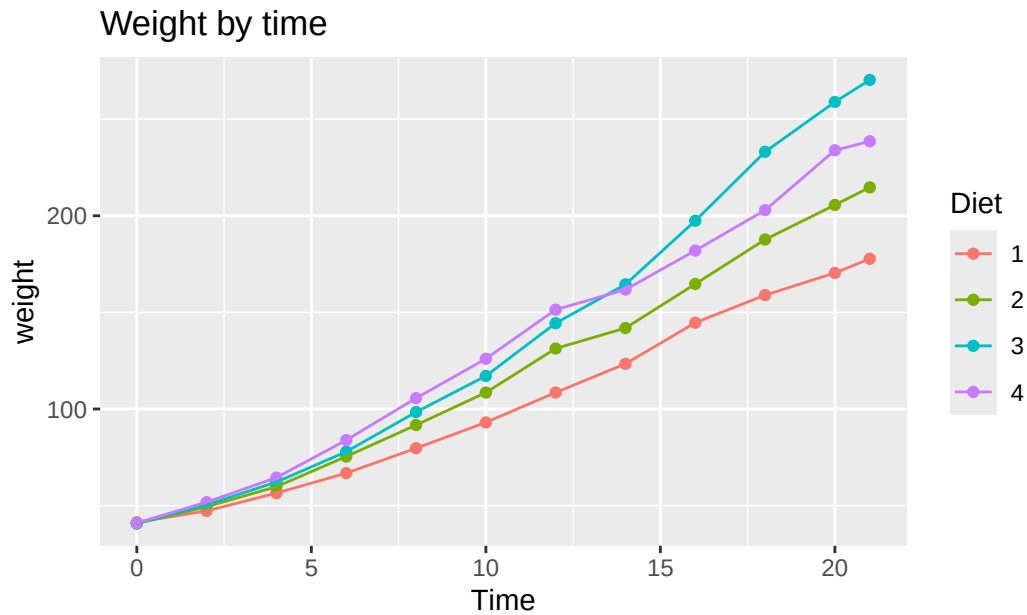
Introduction

In this report, we perform a rudimentary analysis of the built-in ChickWeight Dataset in R.

Table 1: Example rows of the ChickWeight Dataset.

weight	Time	Chick	Diet
42	0	1	1
51	2	1	1
59	4	1	1
64	6	1	1
76	8	1	1
93	10	1	1

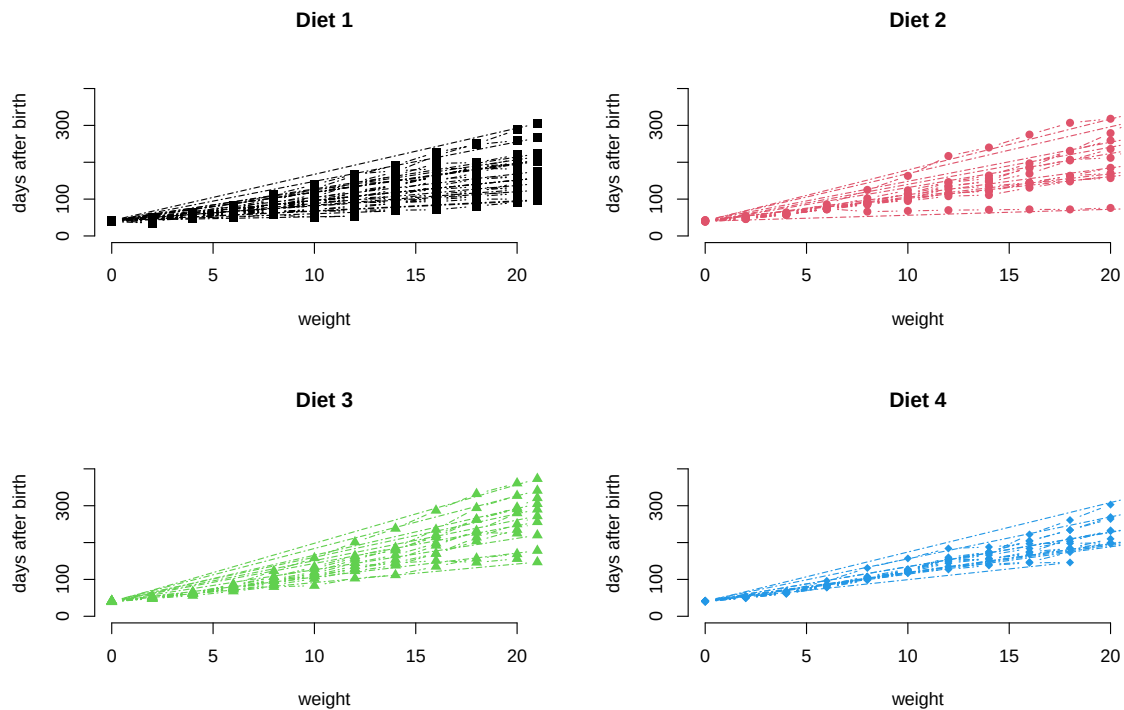
Table 1 give example data entries from the ChickWeight dataset.



Weight by time, averaged over all chicks given the same diet type

```
for (diet in 1:4){
  chicks_level <- chicks[chicks[, "Diet"] == diet,]
  plot(
    main = paste("Diet" , as.character(diet)),
    x = chicks_level$Time,
    y = chicks_level$weight,
    col = diet,
    pch = 14 + diet,
    type = "b",
    lty = 6,
    lwd = 1,
    bty = "n",
    ylim = c(0,450),
    xlab = "weight",
    ylab = "days after birth"
  )
}
```

Example citing: Wickham et al. (2017)



Code appendix

```
knitr::opts_chunk$set(
  collapse = TRUE,
  comment = "#>"
)
include_solutions <- TRUE
library(rmarkdown)
library(knitr)
library(kableExtra)
library(magrittr)
library(ggplot2)
# Put any library imports and other preamble here.

chicks <- ChickWeight
kable(head(chicks))
averaged_chicks <- aggregate(weight ~ Diet + Time , data = chicks, FUN = mean)

#| label: average-chicks-plot
```

```

ggplot(averaged_chicks, aes(x = Time, y = weight, color = Diet, group = Diet)) +
  geom_point() +
  geom_line() +
  labs(title = "Weight by time",
        caption = "Weight by time, averaged over all chicks given the same diet type")

for (diet in 1:4){
  chicks_level <- chicks[chicks[, "Diet"] == diet,]
  plot(
    main = paste("Diet" , as.character(diet)),
    x = chicks_level$Time,
    y = chicks_level$weight,
    col = diet,
    pch = 14 + diet,
    type = "b",
    lty = 6,
    lwd = 1,
    bty = "n",
    ylim = c(0,450),
    xlab = "weight",
    ylab = "days after birth"
  )
}

```

References

Wickham, Hadley, Garrett Grolemond, et al. 2017. *R for Data Science*. Vol. 2. O'reilly Sebastopol.